MIL-S-36971A
3 June 1976
SUPERSEDING
MIL-S-36971(DSA-DM)
23 May 1973

MILITARY SPECIFICATION

SHAKING MACHINE, LABORATORY, 115 volt, 60 Hz, AC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a vibrator type of laboratory shaking machine for operation on 115 volts, 60 Hz, alternating current, and is suitable for accepting and shaking two standard red- or white-corpuscle, blood-diluting pipets.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-packaging, Methods of.
MIL-M-7298	- Manual, Technical: Commercial Equipment.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6640

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, as as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73-11 Dimensions of Plugs and Receptacles, 125 Volts, 15 Amperes, 2-Pole, 3-Wire, Grounding Type, General Purpose.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

UNDERWRITERS' LABORATORIES, INC.

UL 62 Flexible Cord and Fixture Wire

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., NY 11746; 207 East Ohio Street, Chicago, IL 60611; and 1655 Scott Boulevard, Santa Clara, CA 95050.)

3. REQUIREMENTS

3.1 Materials. Materials shall be of the highest quality throughout and shall be suitable in all respects for the purpose intended.

3.2 Style and design. Shaking machine shall be a sturdy, compact, electrical, vibrator-type shaker, capable of producing a homogeneous mixture of blood cells to give uniform blood counts. The shaking machine shall consist essentially of a solenoid assembly, a pipet cradle, controls and a housing. Overall dimensions shall be approximately 3-1/2 inches wide by 6-1/2 inches long by 6-1/4 inches high.

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3.3 Construction.

3.3.1 Housing. The housing shall be molded of high impact-resistant fiberglass reinforced plastic or impact-resistant plastic and shall have approximate overall dimensions of 3 inches wide by 6 inches long by 4-1/2 inches high. The bottom of the housing shall be covered with a stainless steel or aluminum alloy plate, fabricated to conform with the outline of the housing and provided with a 1-1/2 inch diameter ventilation hole centrally located. Four rubber suction cups shall be secured to the bottom of the housing by means of screws. Design, position and size of the cups shall be such that they prevent sliding, tipping or creeping of the shaking machine.

3.3.2 Vibrator assembly. The vibrator assembly shall consist of a cylindrical solenoid coil and a flexible element, to which shall be fastened an armature and a rigid support for the pipet cradle. The center of the coil shall have a cylindrical air core. The armature shall be securely mounted perpendicularly to the flexible element and positioned so as to oscillate freely in one end of the air core of the coil. An axially-adjustable cylindrical shaft shall fit into the other end of the air core of the coil to permit varying the air gap in the magnetic flux path between the armature and the shaft so that the amplitude of the shaking action can be controlled. The amplitude control shall thread through a vibration-friction nut fastened to the coil. The coil shall draw no more than 135 milliamperes at 115 volts, 60 Hz, alternating current. When tested as specified in 4.4.2, the solenoid coil shall show no evidence of electrical breakdown or change in physical characteristics.

3.3.3 Pipet cradle assembly. The pipet cradle shall be molded from an impact-resistant plastic and mounted on the rigid support of the vibrator flexible element. Three loop-shaped supports, fabricated from semi-rigid polyethylene, shall be mounted on the cradle assembly base. These supports shall be positioned on the cradle base to support two pipets. The pipet supports shall permit ready one-hand insertion and removal of the pipet while in motion, yet they shall prevent accidental loss of pipets from the shaking machine when it is adjusted for maximum shaking. The cradle shall be positioned so that fluid leaking out of a pipet will not enter the interior of the shaking machine or will not be trapped in holes or slots.

3.3.4 Power cord and plug. Shall be a flexible, rubber-covered or thermoplastic-covered, 3-wire, 18-gauge, UL-approved cord, not less than 5 feet long outside the housing, that terminates in a 3-prong rubber or thermoplastic plug described in ANSI Standard C73.11. The cord shall be secured to the shaking machine in such a manner as to prevent strain on the terminal connections. The ground lead shall be electrically connected to all exposed metal parts.

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3.3.5 Controls. The shaking machine shall be provided with an "off-on" switch and an amplitude control knob. The amplitude of the shaking action shall be adjustable, with the current on, by means of the amplitude control.

3.4 Performance. Without adjustment of the supports, the shaking machine shall accept one or two 105 to 120 millimeter (mm.) overall length blood-diluting pipets. The shaking action of the pipet shall be a combination of rotation about the long axis of the pipet, movement in both directions along its long axis, motion at right angles to its long axis, and tilt of its long axis. Sufficient amplitude and speed of shaking shall be developed so that the pipets are bounced clear of the support and rotate independently of the pipet supports. When tested as specified in 4.4.1, the shaking machine shall operate without any type of failure, without tendency for components to be loosened by vibration, and without tendency for shaking machine to slide, creep or tip.

3.5 Finish. All exposed metal parts shall be coated to resist corrosion.

3.6 Identification marking. In addition to commercial markings, the NSN and contract or purchase order number shall be legibly and permanently marked. The additional markings may be provided on labels secured by adhesives.

3.7 Service data. The contractor shall furnish with each unit two copies each of a technical manual which contain complete instructions for installation, operation and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. Two preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish

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manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.8 Workmanship. Workmanship shall be first class throughout. The shaking machine shall be free from defects which detract from its appearance or may impair its serviceability. The shaking machine shall efficiently perform its functions in a manner suitable for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

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* TABLE I. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Acceptance number shall be zero for all sample sizes.

TABLE II. Sampling for tests

Component (unit of product)	Characteristic	Requirement	Test paragraph	Inspection level
End item	Operation	3.4	4.4.1	100%
End item	Solenoid breakdown	3.3.2	4.4.2	S-1

4.3 Examination. The shaking machine shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following table:

TABLE III. Classification of defects

Categories	Defects*
Major	
101	Housing not free of cracks.
102	Cradle assembly not free of cracks.
103	"Off-on" switch missing or inoperative.
104	Amplitude control missing or inoperative.
105	Metal parts not free of corrosion.
106	Electrical cord not secured to shaking machine.
107	Pipet cradle assembly not securely mounted.
Minor	
201	Identification plate not furnished.
202	Printing on identification marking not clear or legible.
203	Service data not furnished.
204	Housing not free of scratches or other surface imperfections.

* Examination shall not be restricted to the classified possible defects listed.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Operation. The shaking machine shall be loaded with two blood-diluting pipets. The amplitude control shall be set for minimum amplitude. With the line plug connected, the switch shall be set to the "on" position, and the amplitude control shall be slowly increased to maximum agitation, which shall be maintained for one minute. The amplitude control shall then be gradually decreased to the minimum setting, and the switch shall be set to the "off" position.

4.4.2 Solenoid breakdown. The shaking machine shall be loaded with two blood-diluting pipets filled with water. The shaking machine shall be operated continuously for 48 hours within the upper range of the amplitude control settings.

* 4.5 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the Group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

* 5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C, as specified (see 6.1).

5.1.1 Unit of issue. Each shaking machine, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Shaking machines shall be cleaned in accordance with MIL-P-116, method C-1.

5.1.2.2 Drying. Shaking machines shall be dried in accordance with MIL-P-116.

5.1.2.3 Preservative application. Preservatives shall not be used with the exception of lubrication applied to internal moving parts as deemed necessary by the contractor.

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5.1.2.4 Unit packaging. Each unit shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.3 Level C. Shaking machine shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. Twenty four units, packaged as specified in 5.1.2, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Twenty four units, packaged as specified in 5.1.2, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The packaged shaking machine shall be packed in a shipping container in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking. Unit package and exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement document should specify the following:

a. National Stock Number (NSN).
b. Title, number and date of this specification.
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet Military requirements.

6.3 This specification covers the following item shown in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-299-8693	SHAKING MACHINE, LABORATORY, 115 volt, 60 HZ, AC

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0948

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-36971A SHAKING MACHINE, LABORATORY, 115 volt, 60 Hz, AC

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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